

# SwadeshAI: Building India’s AI Sovereignty through Fundamental Research

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## Abstract

India is the world’s largest mobile user base for ChatGPT and the third-largest for DeepSeek, yet contributes between 0.1 and 1% of accepted papers at top artificial intelligence (AI) conferences and has produced no frontier-class AI architectures or foundation models. The Stanford AI Index 2026 records India at 7.6% of global AI publications but absent from both the notable-models and high-impact-patents rankings dominated by the United States and China. India’s Rs.10,372 Cr IndiaAI Mission allocates nothing to fundamental AI research in theory, algorithms, or mathematical foundations. This paper argues that AI sovereignty—the capacity to build, deploy, and govern indigenous AI capabilities—cannot be achieved through an application-and-services strategy alone. Drawing on bibliometric evidence from the Stanford AI Index, Scopus, and the IndiaML tracker, as well as comparative analysis of China’s trajectory from AI follower to near-parity with the United States, we show that foundational research capability is the binding constraint on India’s AI self-reliance story. To remedy this situation, we propose a set of policy interventions: earmarking a dedicated AI research fund within the Anusandhan National Research Foundation (ANRF), redirecting a fraction of India’s mandatory corporate social responsibility (CSR) expenditure through the existing Schedule VII mechanism of the Companies Act, and reforming UGC and AICTE promotion criteria from publication count to venue quality. Together, these measures can substantially expand India’s frontier AI research capacity without waiting for new budgetary allocations.

**Keywords:** Atmanirbhar Bharat; Sovereign AI; artificial intelligence; R&D policy; fundamental research

## 1. Introduction

India’s relationship with artificial intelligence (AI) is defined by a striking paradox. The country is the world’s largest market for ChatGPT’s mobile application and the third-largest user base for DeepSeek [23]. India’s AI market is projected to reach \$8 billion by 2025, growing at 40% compound annual growth rate [22]. Indic-language AI represents a massive latent opportunity. India constitutes 18% of the global population, yet Indic languages comprise merely 1% of the Common Crawl corpora [10] that is extensively used in training frontier large language models (LLMs). The demographic and market opportunity for AI leadership could hardly be more

favourable, and the moment for decisive action has arrived.

Data curated by the Stanford AI Index [22, 23] paint a bleak picture of AI R&D in India. India produced zero of the “notable AI models” tracked by the Stanford AI Index, while the United States produced 59 and China 35 [23, Figure 1.1.1]. India contributes between 0.1 and 1% of accepted papers at leading AI conferences [2, 8]. India’s share of global AI patents stands at 0.4%, compared to China’s 74.2% and the United States’ 12.1% [23, Figure 1.7.2]. Among the list of countries contributing to the 100 most-cited AI publications in 2021–2024, India is at the last place with 1-4 publications per year, with small nations like Hong Kong, Switzerland, and Singapore appearing higher [23, Figure 1.6.12].

This gap between consumption and creation is not merely a matter of academic prestige. It represents a structural deficit in achieving true AI *sovereignty*: the capacity of a nation to shape, not merely use, the foundational technologies that increasingly determine economic competitiveness, security capability, and geopolitical influence. One can argue that the *swadeshi* aspiration embedded in the *Atmanirbhar Bharat* vision—building at home rather than importing—is precisely what is absent in India’s AI strategy, which imports the intellectual foundations and indigenises only the surface layer. India’s Rs.10,372 Cr IndiaAI Mission reflects this orientation. It devotes the overwhelming majority of its budget to compute infrastructure, startup financing, and application development, with no explicit allocation for curiosity-driven basic research from which sovereign capability is built [4, 15].

The paper is structured as follows. Section 2 presents bibliometric evidence mapping India’s AI research deficit. Section 3 diagnoses the structural root causes. Section 4 examines China’s trajectory as a comparative counterfactual, and draws on South Korea’s success story of R&D-driven growth. Section 5 proposes policy recommendations to make progress on our broad vision, and Section 6 offers a conclusive discussion.

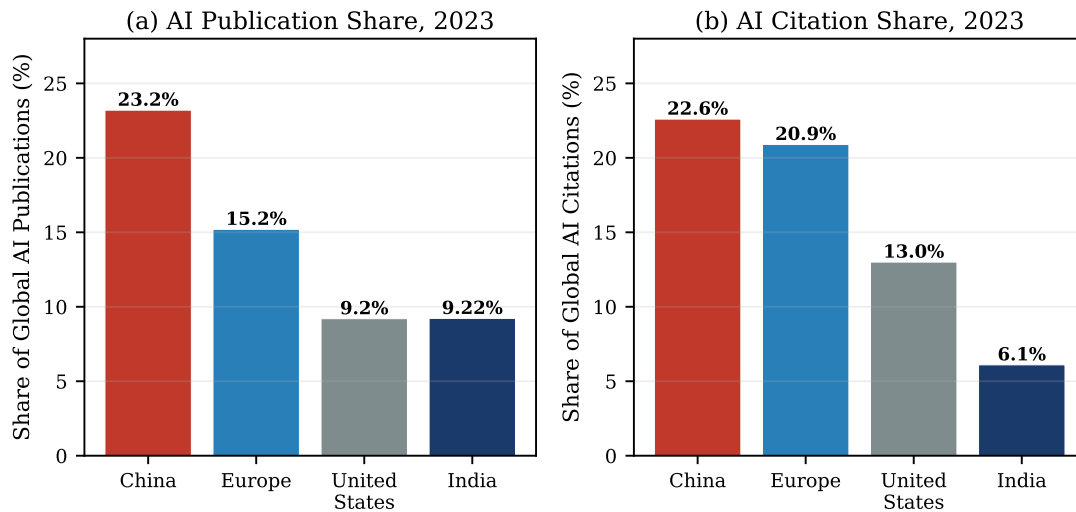
## **2. Mapping the Deficit: India’s AI Research in Global Context**

Drawing on the Stanford AI Index reports for 2025 [22] and 2026 [23], we assess India’s position in the global AI research landscape along three dimensions: the volume and impact of its scholarly publications, its representation at top AI research conferences, and its capacity to produce frontier AI models. Across all three, the same pattern emerges: impressive surface-level metrics that dissolve on contact with quality-adjusted comparisons.

### **2.1. Volume without impact**

India’s research output in AI, measured by publication volume, appears respectable (Figure 1 left panel). India accounted for 9.22% of global AI publications in 2023, roughly on par with the United States at 9.2%, though well behind China at 23.2% [22]. In the broader Scopus database, Indian researchers contributed more than 16 Lakh papers between January 2020 and May 2025: the highest volume in the world [1].

However, this volume masks a deeper quality deficit. When the impact of published papers in AI is considered, measured in citations generated, India trails substantially (Figure 1 right panel). Chinese AI publications accounted for 22.6% of all AI citations in 2023, followed by Europe at 20.9% and the United States at 13.0%; India’s share was 6.1% [22]. India’s H-index of 925 is 71.2% lower than the United States’ H-index of 3,213 [1]. Of the roughly 200 Indian journals indexed in Scopus, only 12 are classified as Quartile 1 (top 25%), with the vast majority sitting in Quartiles 3 and 4 [1].



**Figure 1:** AI publication versus citation shares by country/region, 2023. *Source: Stanford HAI [22], Figures 1.1.6–1.1.7.*

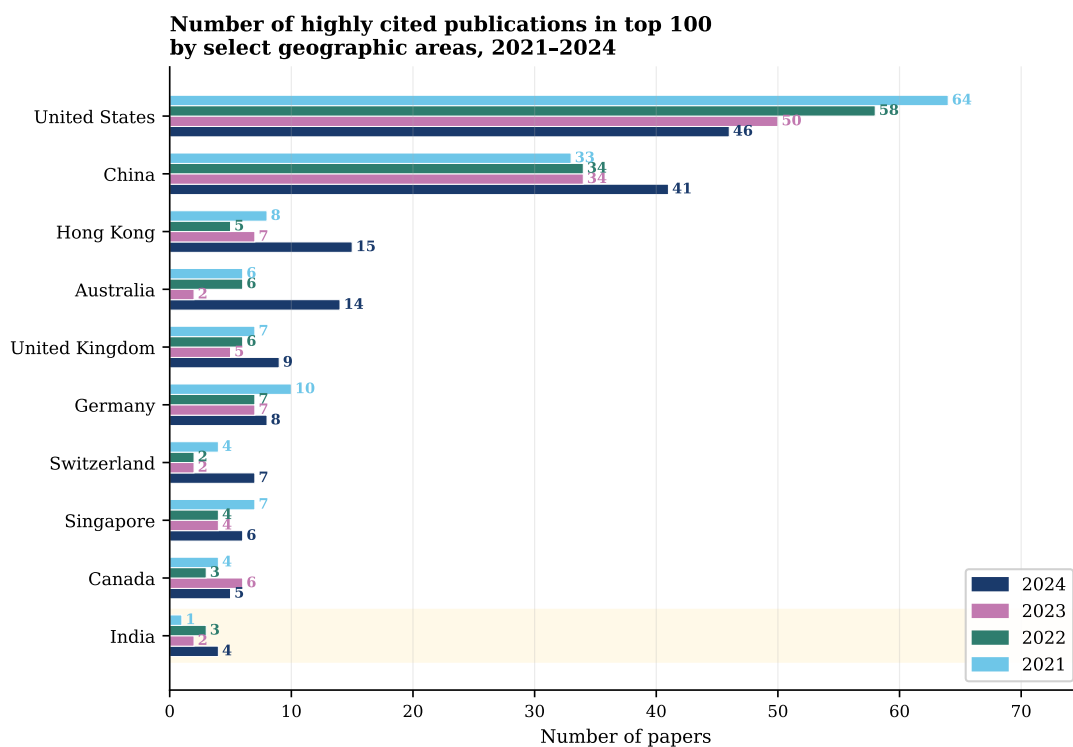
Looking at how things changed over time, between 2013 and 2023, China’s publication share rose from approximately 10% to 23.2% whilst its citation share grew commensurately from 8% to 22.6%. In comparison, India’s publication share grew from approximately 5% to 9.2% over the same period, but its citation share grew far more slowly, from approximately 3% to 6.1% [22]. India’s papers are produced at increasing volume but are not read, cited, or built upon at a commensurate rate. This indicates a research ecosystem that is optimised for production rather than influence.

## 2.2. The top-conference gap

In AI, unlike most scientific disciplines, peer-reviewed conferences rather than journals serve as the primary venue for disseminating frontier research. The IndiaML tracker, an open-source initiative tracking Indian-affiliated papers at major machine learning venues using the OpenReview API, documents that India’s contribution to accepted papers at four top AI conferences (NeurIPS, ICML, ICLR, and CVPR) ranges between 0.1 and 1% [2, 8]. This is an extraordinarily low figure for a country with the world’s largest developer workforce and the largest consumer market for AI platforms.

The most revealing metric, however, is the distribution of the 100 most-cited AI publications. In

2024, the United States contributed 46 papers, China 41, Hong Kong 15, Australia 14, and the United Kingdom 9. India contributed 4. This pattern holds across the previous three years for which data are available. Overall, India lies far behind others in top-100 AI paper counts across 2021–2024 [22]. Figure 2 presents this data. The implications extend well beyond bibliometric standing. Top-cited papers introduce the architectures, training methods, and theoretical results that subsequent work builds upon; they define what the field studies and what counts as progress. A country absent from this list cedes the power to set that direction to others. Moreover, when India’s AI researchers work on problems framed by American and Chinese researchers, using methods designed for deployment contexts quite different from India’s own, the result is a body of work that solves the wrong problems at scale. As a consequence, the unique challenges Indian AI users face, such as lack of network connectivity, availability on-the-edge, and multilingual and code-switched usage remain unaddressed or suboptimally addressed.

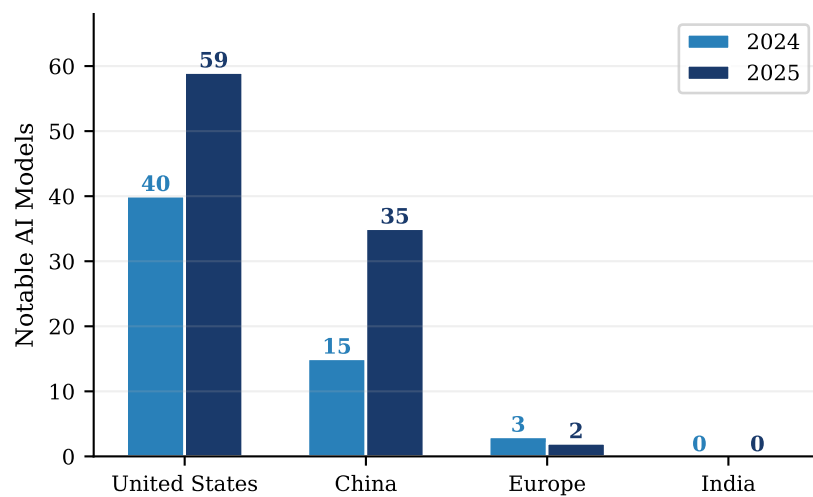


**Figure 2:** Number of papers in the top-100 most-cited AI publications, by country, 2021–2024. *Source:* Stanford HAI [23, Fig. 1.6.12].

### 2.3. The frontier-models gap

The Stanford AI Index tracks “notable AI models” representing significant technical advances. In 2024, the United States produced 40 such models, China 15, and Europe 3; in 2025, the United States produced 59 and China 35. India produced zero in both years [22, 23]. Indian foundation models do exist: Krutrim (Ola, trained on 2 trillion tokens across 10+ Indian languages), the Param series (BharatGen/IIT Bombay consortium, up to 17 billion parameters in 22 languages), and the Sarvam models launched under the India AI Impact Summit 2026 (30B and 105B

parameter variants pretrained from scratch on large Indic corpora). Among them, the Sarvam models are the most substantial contribution, since pretraining from scratch on Indic data requires significant compute and data-curation effort, and signals a maturing of indigenous model-building capacity [10]. Yet even here, the ceiling is clear: none of these models appear on global leaderboards, and none has produced methodological innovations such a new architecture, training objective, or algorithmic technique comparable to those emerging from American and Chinese laboratories (a point we develop in Section 4). Building a pretrained Indic model is a necessary step toward AI sovereignty; it is not the same as advancing the science of building and understanding AI models. A nation with genuine AI sovereignty should ideally progress both. Figure 3 illustrates this frontier-models gap, and Table 1 summarises the full consumer-creator asymmetry.



**Figure 3:** Notable AI models produced by country of origin, 2024–2025. India produced zero notable models in both years. AI sovereignty requires the capacity to produce frontier models, not merely to localise them. *Source:* Stanford HAI [22, 23].

**Table 1:** India’s AI consumer–creator paradox: key metrics across dimensions of AI sovereignty.

| Metric                             | India          | China                | United States  |
|------------------------------------|----------------|----------------------|----------------|
| AI publications (% global, 2023)   | 9.22%          | 23.20%               | 9.20%          |
| AI citations (% global, 2023)      | 6.10%          | 22.60%               | 13.03%         |
| Top-100 cited AI papers (2024)     | <b>4</b>       | 41                   | 46             |
| Notable AI models (2025)           | <b>0</b>       | 35                   | 59             |
| AI patents (% world total, 2023)   | 0.4%           | 74.24%               | 12.06%         |
| Top-conference papers (% accepted) | 0.1–1%         | ~30–40%              | ~30–40%        |
| Private AI investment (2025)       | \$1.25B (govt) | \$12.4B              | \$285.9B       |
| GERD/GDP                           | 0.64%          | 2.68%                | 3.59%          |
| Frontier foundation models         | None           | DeepSeek-R1,<br>Qwen | GPT-4o, Claude |
| ChatGPT mobile user base           | <b>Largest</b> | —                    | —              |
| DeepSeek user base                 | 3rd largest    | 1st                  | —              |
| Semiconductor design workforce     | 20%            | —                    | —              |
| Semiconductor fabrication capacity | 0.5%           | —                    | —              |

*Sources:* Basak and Chopra [2], Endiya Partners [5], Indian Society of Artificial Intelligence and Law [8], Kallappa et al. [10], Ministry of Electronics and Information Technology [15], Organisation for Economic Co-operation and Development [19], Stanford HAI [22, 23]. Bold entries highlight the sharpest dimensions of the consumer-creator asymmetry.

### 3. Diagnosing the Root Cause

Three factors explain why India’s AI ecosystem has taken the shape it has: the composition of its flagship national AI investment, a misaligned academic incentive system, and a deeply entrenched services-over-research orientation in the private sector. Each is self-reinforcing, and addressing any one of them in isolation will be insufficient without simultaneous reform of the others.

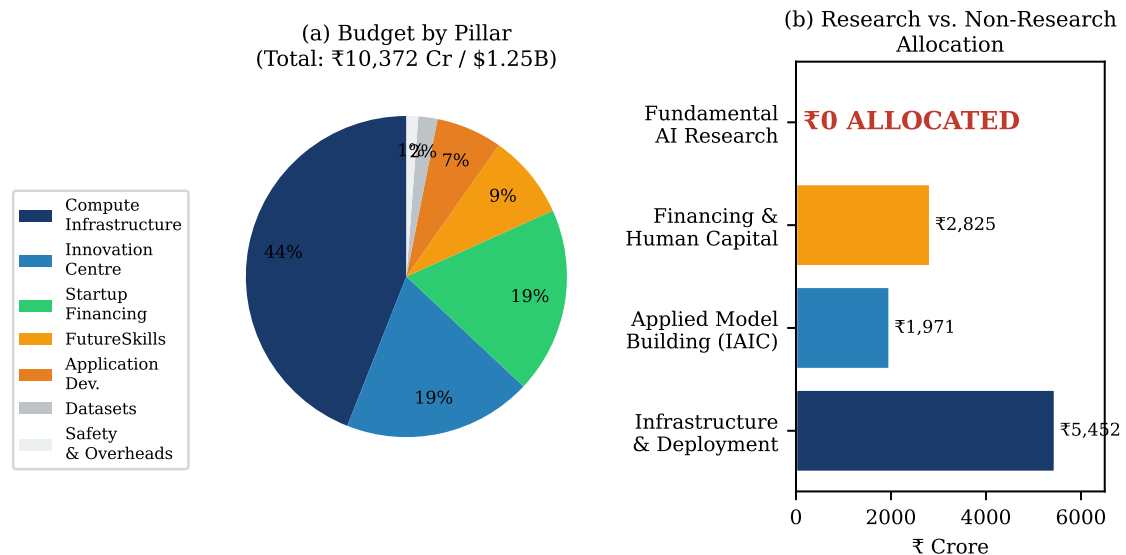
#### 3.1. The IndiaAI Mission’s architecture

The IndiaAI Mission, approved by the Cabinet in March 2024 with a five-year budget of Rs.10,372 Cr (approximately USD \$1.25 Billion), is India’s flagship national AI initiative [15]. Its seven pillars and their allocations are: Compute Capacity (Rs.4,563 Cr, 44%), Innovation Centre (Rs.1,971 Cr, 19%), Startup Financing (Rs.1,943 Cr, 18.7%), Future Skills (Rs.883 Cr, 8.5%), Application Development (Rs.689 Cr, 6.6%), Datasets Platform (Rs.200 Cr, 1.9%), and Safe & Trusted AI (Rs.20 Cr, 0.2%) [4] (Figure 4).

The budget reveals a structural bias towards deployment over discovery. Nearly two-thirds of the allocation funds hardware and capital access. The Innovation Centre<sup>1</sup>—the pillar supposedly oriented towards research—is tasked with developing “indigenous Large Multimodal Models

<sup>1</sup><https://indiaai.gov.in/hub/indiaai-innovation-centre>

and domain-specific models” [15], which in practice means applied model-building, not fundamental research in learning theory, optimisation and algorithms, or statistical machine learning. Importantly, no pillar is explicitly dedicated to curiosity-driven basic AI research.



**Figure 4:** IndiaAI Mission budget breakdown by pillar (left) and reframed as research versus non-research allocation (right). No funding is explicitly allocated to fundamental AI research. A mission structured entirely around deployment, applied model-building, and financing cannot deliver AI sovereignty: it produces AI users, not AI inventors. *Source:* Parliamentary response, December 2024 [4]; Cabinet approval [15].

A sovereignty agenda requires investment commensurate with external benchmarks in similar category. Looking from this comparative perspective compounds the above structural concern. The IndiaAI budget of USD \$1.25 billion over five years translates to approximately USD \$250 million per year. In comparison, private investment in AI in the United States in 2025 alone was USD \$285.9 billion, roughly 1,140 times India’s annual government allocation [23]. Even within Asia, China’s government semiconductor fund alone is \$47.5 billion, 38 times India’s entire five-year AI budget [22].

### 3.2. Misaligned incentives

India’s academic promotion criteria, governed by the University Grants Commission (UGC) and All India Council for Technical Education (AICTE) regulations, reward publication count in Scopus- or Web of Science-indexed outlets rather than the quality of the publication venue or citation impact. This creates a rational incentive to publish in Q3 and Q4 journals, where acceptance is easier, rather than attempt the high-risk, high-reward submissions in world-class venues with much lower acceptance rates. Between 2020 and 2025, India contributed 1.6 million Scopus papers, the most in the world, but only 12 Indian journals achieve Q1 status [1]. A 2025 study of the top 1% of highly cited India-affiliated papers found that papers with more than 90% self-citations are predominantly domestic, small-team, and recent—suggesting that citation

inflation, rather than genuine impact, is being incentivised [11].

This misalignment has a direct consequence for AI research talent. A faculty member at an Indian institution faces no career incentive to spend two years developing a theoretically novel contribution and submitting to a top conference like NeurIPS, where rejection is probable and the timeline uncertain, when five Q4 journal papers produce the same promotion outcome in the same period. The incentive architecture systematically directs India’s most capable AI researchers away from frontier research that would build the foundations of AI sovereignty.

### **3.3. The IT-services comfort zone**

India’s technology ecosystem, built around major consulting and services companies, is fundamentally configured for execution rather than invention. These organizations are designed and built around the risk-averse strategy of simply deploying already mature technologies without investing in forward-looking research. As a contextual example in the domain of AI, they are consumers of open-source libraries like PyTorch and TensorFlow, without contributing substantially to either the algorithmic foundations or their worldwide development efforts. The corporate R&D culture exemplified by global success stories such as AT&T Bell Labs and IBM Research in the USA, Huawei Noah Lab and ByteDance’s AI Lab in China, or Samsung Advanced Institute of Technology (SAIT) in Korea has no Indian equivalent at scale.

This pattern of downstream capability building without upstream research is self-reinforcing and structurally inimical to AI sovereignty. Indian IT firms, having built profitable businesses around service delivery, face no competitive pressure to invest in fundamental research. Their clients—predominantly American and European firms—perform the upstream R&D themselves. As AI increasingly commoditises the deployment layer through APIs, pretrained models, and low-code automation frameworks, the value will shift further upstream to those who create the foundational technologies. India’s current positioning leaves it exposed to precisely this shift.

The semiconductor story illustrates the stakes. India accounts for 20% of the global semiconductor design workforce, approximately 1.5 Lakh professionals, but contributes only 0.5% to global fabrication capacity [5]. Design-without-fabrication is the hardware-layer analogue of India’s AI situation: downstream capability without the sovereign control that comes only from owning the foundational layer.

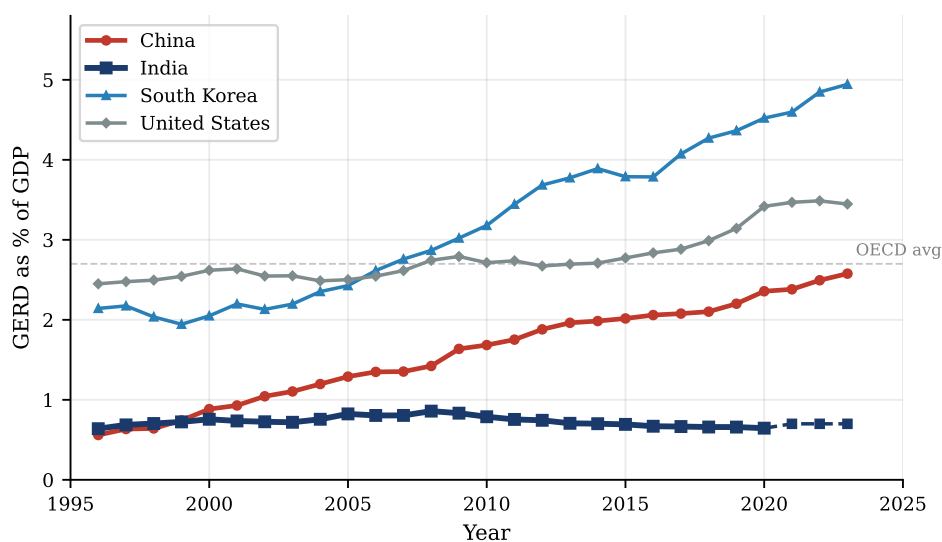
## **4. Success Stories: How Fundamental Research can Build Self-reliance**

Recent and relevant progress in two countries, China and South Korea, offer instructive counterpoints to India’s trajectory. Both began from positions of technological followership broadly comparable to India’s today, and both made deliberate, sustained investments in basic research as a matter of explicit national policy. Their outcomes demonstrate that the pipeline from foundational research to applied capability to sovereign competitiveness is real, reproducible, and historically achievable within a generation.



## 4.1. China

China’s transformation from a follower to a near-peer of the United States in broad R&D provides the most relevant comparative lesson for India (Figure 5). In 1996, China’s gross expenditure on R&D (GERD) as a percentage of GDP was 0.56%, comparable to India’s 0.76% at the time [24]. By 2023, China had nearly quintupled this ratio to 2.58%, with the National Bureau of Statistics reporting a further increase to 2.68% by 2024 [16]; India’s ratio, meanwhile, peaked at 0.86% in 2008 before declining, with the most recent official figure standing at 0.64% for 2020–21 and parliamentary responses indicating it has remained near 0.7% since [3, 20]. China’s total R&D expenditure in 2024 exceeded 3.6 trillion yuan (approximately USD \$500 billion), and on a purchasing-power-parity basis China surpassed the United States for the first time in absolute R&D spending, at USD \$1.03 trillion versus the USA’s USD \$1.01 trillion [12, 19].



**Figure 5:** Gross expenditure on R&D as a percentage of GDP, 1996–2023. China rose substantially from 0.56% to 2.58% whilst India’s trajectory, which peaked at 0.86% in 2008, declined to 0.65% by 2020. South Korea, at 4.94%, leads the world. The divergence between China and India exemplifies two nations that started from the same point and made fundamentally different choices about research investment: choices whose consequences are now visible in AI sovereignty outcomes. India’s dashed segment (2021–2023) is carried forward at 0.7%, the upper bound cited in parliamentary responses, as DST has not yet published official GERD data beyond 2020–21. *Sources:* Department of Science and Technology [3], National Bureau of Statistics of China [16], Organisation for Economic Co-operation and Development [19], Press Information Bureau [20], World Bank [24].

In AI research outputs, China went from under 5% of global AI publications in 2000 to 36% by 2025, emerging as the single most dominant contributor [7]. China now leads in both AI publication volume (23.2% in 2023) and citation share (22.6%), and has recently overtaken even in high-impact publications, challenging the assumption that Western powers retain dominance in influential AI scholarship [7, 22]. In comparison, analysis of accepted papers at NeurIPS, ICML, and ICLR using the IndiaML tracker consistently places Chinese-affiliated authors

among the top contributors by country, whilst Indian-affiliated authors account for under 1% of accepted papers [2, 8].

China’s AI research build-up was embedded in a broader basic-research expansion. In 2024, China’s investment in basic research grew 10.7% year-on-year to 250.1 billion Yuan, representing 6.9% of total R&D expenditure [17]. University capacity massively expanded in parallel: the number of higher education institutions grew from approximately 1,000 to over 3,000 between 2000 and 2024. Today, Chinese institutions hold 7 of the top 10 positions in the Nature Index [18]. The Chinese Academy of Sciences alone, with a Nature Index Share of 2,243, outproduces all of India combined, which has a nature index share of approximately 1,494 [18].

The DeepSeek story distils this lesson. A relatively small Chinese team, operating under US chip export controls that restricted access to advanced GPUs, produced DeepSeek R1, an open-source LLM that matched frontier American proprietary systems. This was possible because the algorithmic innovation behind DeepSeek R1—an efficient neural network architecture and a novel reinforcement learning algorithm—did not require frontier hardware. Instead, it was made possible by the investment on frontier research [23]. DeepSeek demonstrates that AI sovereignty is primarily an intellectual achievement, not a hardware or capital one. India’s response to chip export controls, or to any future geopolitical disruption in AI supply chains, will depend entirely on whether it has built the research base to innovate around constraints. Until now, it has not.

China also implemented an explicit institutional strategy to build presence in top AI conferences as a component of AI sovereignty. Universities such as Tsinghua, Peking, and Zhejiang were incentivised and funded to target publications at NeurIPS, ICML, and CVPR. Faculty hiring and promotion increasingly weighted top-venue publications highly. By 2023, Tsinghua University alone tied with Google for the most papers in the top-100 most-cited AI list, with eight each [22]. In comparison, no Indian institution appears in this ranking. The lesson here is that building R&D visibility at the world stage presence requires institutional strategy and incentive alignment, not merely researcher talent—which India possesses in abundance.

#### **4.2. South Korea**

South Korea’s R&D intensity, at 5.2% of GDP, is the highest in the world [19]. This investment transformed the country from a labour-intensive exporter of textiles and footwear in the 1960s to the global leader in semiconductors, displays, and batteries. The transformation was *deliberately* research-led: the Korea Institute of Science and Technology (KIST) and the Korea Advanced Institute of Science and Technology (KAIST) were founded (in 1966 and 1971, respectively) as national R&D anchors with explicit mandates for basic research. Samsung’s corporate R&D centre (SAIT) was restructured from a development-oriented laboratory to a basic-research institution to enable the firm’s transition from fast-follower to sovereign innovator [6]. In AI specifically, South Korea now leads the world in AI patents per capita at 17.3 per 1 Lakh inhabitants—a concrete measure of AI sovereignty in the intellectual-property dimension [22].

South Korea’s experience also illuminates the danger of the “fast-follower” model that India’s IT-services sector exemplifies. Korean policymakers recognised by the mid-1990s that the fast-follower strategy had reached its limits: as followers caught up, profitability was competed away, and the only escape was to move up the value chain through basic-research investment [6]. India’s IT sector faces the same structural pressure today, accelerated by rapid AI-driven commoditisation of software development and service-delivery tasks. The question is whether India will make the sustained, sovereignty-oriented research commitment South Korea made before its window of opportunity closes.

## **5. Recommendations for Building AI Sovereignty from the Research Up**

The policy proposals below are sequenced to address the concerns identified in Section 3, rather than their symptoms. Together, they amount to a coherent programme for shifting India’s AI investment from the surface layer to the foundational layer.

### **5.1. Create a dedicated AI Research Fund**

The Anusandhan National Research Foundation (ANRF), established in 2023, has been set up to foster impact-oriented research in the model of the United States National Science foundation (NSF). While the ANRF Advanced Research Grant (ARG) programme provides generous support for substantial R&D projects, and the ARG MATRICS programme supports basic research in mathematics, theoretical sciences, and quantitative social sciences, there is no explicit support for basic research in AI. We recommend that ANRF should earmark at least Rs.2,000 Cr over five years specifically for investigator-initiated, curiosity-driven AI research, encompassing learning theory, optimisation and algorithms, statistical machine learning, and the foundations of trustworthy AI. This fund should be modelled on the NSF’s Computing and Communication Foundations (CCF) programme, with rigorous international peer review at the level of NeurIPS and ICML programme committees, instead of UGC panels. No requirement for immediate application should be imposed. The objective of this programme would be to build the long-term intellectual foundations from which applied innovations, and sovereign capability, would emerge. At approximately USD \$210 million over five years, the fund is a fraction of what a single frontier-class algorithmic innovation is worth in market terms, yet it would represent India’s first dedicated investment in foundational AI knowledge.

### **5.2. Reform academic incentives**

UGC and AICTE promotion criteria in AI-relevant disciplines (e.g. Computer Science, Statistics, Data Science) should replace publication-count metrics with quality-weighted indicators. A single paper accepted at NeurIPS, ICML, or ICLR should carry more weight in promotion and tenure decisions than ten papers in Q4 journals. Field-weighted citation impact and top-decile publication share should be incorporated into assessment frameworks. This single reform would redirect thousands of India’s AI researchers from volume-chasing to frontier-seeking, aligning individual incentives with the national AI sovereignty agenda. The cost to do so is negligible,

but the potential impact is transformational.

### **5.3. Build AI research institutes, not just data centres**

The IndiaAI Innovation Centre should be restructured as a network of five to ten AI research institutes, housed at institutions such as IISc, select IITs, ISI, CMI, TIFR, and the recently set up Indian AI Research Organization (IAIRO), with protected funding for fundamental research, competitive international salaries, and a mandate to publish at top venues rather than to build products. These institutes should be modelled on MILA (Montréal), the Canadian Institute for Advanced Research (CIFAR), or the Max Planck Institutes in Germany—all of which demonstrate that sustained public investment in basic research generates outsized returns in both knowledge and sovereign capability. To provide funds to support this long-term value creation, a major share of IndiaAI funding should be allocated to basic research instead of on startups or GPU clusters. This would endow the top institutions of the country with the freedom to pursue foundational questions, the ability to retain the best research talent by incentivising them to pursue world-class research *in India*, and the patient resilience against hardware constraints, sanctions, or supply-chain disruptions to pursue frontier research.

### **5.4. Close the Indic data gap through research**

The 1% representation of Indic languages in Common Crawl is not simply a data problem. It is a research problem in low-resource NLP, with interesting technical directions like cross-lingual transfer learning, code-switching modelling, and script-diverse tokenisation [10]. Funding data-centric AI projects as fundamental research programmes, rather than treating them solely as dataset-creation exercises, would simultaneously address India’s linguistic AI gap and position India as a global leader in inclusive AI research. Sovereignty in AI for India requires AI that actually works for India’s population: this cannot be achieved by fine-tuning English-trained models, only by building the research foundations of multilingual intelligence.

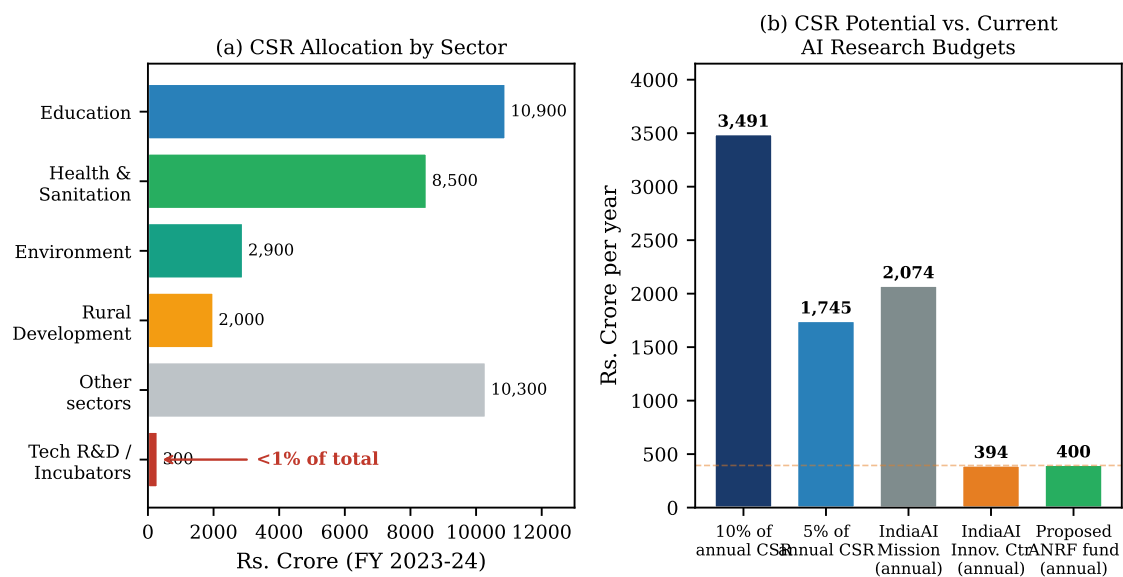
### **5.5. Redirect corporate social responsibility funds towards AI research**

India possesses a financing instrument for AI sovereignty that is large, recurring, legally established, and almost entirely untapped: mandatory corporate social responsibility (CSR) expenditure. Under Section 135 of the Companies Act 2013, every company with a net worth of Rs.500 Cr, turnover of Rs.1,000 Cr, or net profit of Rs.5 Cr must spend at least 2% of its average net profit on CSR activities. This is the world’s first such statutory mandate, and the sums involved are now substantial: total CSR expenditure reached a record Rs.34,909 Cr in FY 2023–24, having tripled over the preceding decade, with private companies contributing roughly 84% [14, 21].

The legal mechanism to channel these funds into research already exists. Following amendments to Schedule VII (Item ix) of the Companies Act in 2019 and August 2020, CSR contributions to incubators and to R&D projects in science, technology, engineering, and medicine—when routed through publicly funded universities, the IITs, national laboratories, or agencies such as

DST, CSIR, and the Ministry of Electronics and Information Technology (MeitY)—are explicitly eligible CSR activities [9, 13]. In other words, an Indian company can already discharge its statutory CSR obligation by funding fundamental AI research at an IIT or an IndiaAI-affiliated institution.

Yet this channel is barely used. Contributions to technology incubators and R&D accounted for less than 0.2% of total CSR spending between 2014–15 and 2017–18, and the category continues to receive the smallest share of any in Schedule VII [9]. The overwhelming majority of CSR funds flow to education and healthcare, which together absorb more than 55% of the total [14]. These are worthy causes, but the concentration reflects institutional habit and reputational convenience rather than strategic national priority. Figure 6 presents the sectoral allocation of CSR funds, alongside the scale of the opportunity.



**Figure 6:** (a) CSR allocation by sector, FY 2023–24 (approximate; technology and R&D receive under 1%). (b) What a modest reallocation of CSR could fund, compared with current AI research budgets (annualised). Even 10% of annual CSR (Rs.3,491 Cr) exceeds the entire IndiaAI Mission’s average annual outlay and is roughly nine times the annual allocation of its Innovation Centre. *Sources:* Elets [4], Invest India [9], Ministry of Corporate Affairs [14], Ministry of Electronics and Information Technology [15], PRIME Database Group [21].

Redirecting just 10% of annual CSR expenditure towards AI research would mobilise approximately Rs.3,491 Cr per year: a sum that exceeds the entire IndiaAI Mission’s average annual budget of Rs.2,074 Cr and is roughly nine times the Rs.394 Cr annual allocation of its Innovation Centre. Even a conservative 5% reallocation (Rs.1,745 Cr per year) would more than quadruple India’s effective annual public investment in applied AI research. This requires no new taxation and no new government expenditure: the funds are already being collected and spent under existing law.

Three policy measures would unlock this channel. First, the Ministry of Corporate Affairs

should issue clarificatory guidance establishing a streamlined, pre-approved roster of AI research institutions and projects—removing the ambiguity that has historically deterred companies from the Schedule VII research route [9]. Second, the IndiaAI Mission’s Innovation Centre should be empowered to receive and pool CSR contributions, matching them against government funds to amplify their impact. Third, the government should consider a modest mandate or strong incentive—for instance, encouraging technology-sector firms above a threshold to direct a defined fraction of their CSR towards STEM research—recognising that a research-capable India is itself a public good of the highest order. For a country whose technology firms have profited enormously from the global AI value chain whilst contributing little to its foundations, redirecting a fraction of mandatory social spending towards building indigenous research capacity is a fitting expression of corporate responsibility in the age of AI.

### **5.6. Incentivise private-sector AI R&D at the frontier**

Beyond CSR, India’s private sector contributes only 36% of national R&D expenditure, compared to over 70% in the United States, China, and South Korea [20]. Matched-funding schemes, in which government co-invests in corporate fundamental AI research rather than product development, should be established. Production-linked incentive (PLI) beneficiary firms in AI and electronics should be required to maintain a minimum R&D-to-revenue ratio and to publish at peer-reviewed venues. The long-term objective would be to cultivate a corporate research culture, at par with previously mentioned global standards such as AT&T Bell Labs and Samsung’s SAIT, that India currently lacks and that AI sovereignty ultimately requires.

### **5.7. Invest in research talent as a sovereignty asset**

PhD fellowships in AI and related mathematical sciences should be dramatically expanded in amount and coverage. Stipends should be raised to internationally competitive levels, and more postdoctoral positions should be created within Indian institutions. Reverse brain-drain programmes should target Indian-origin researchers in frontier AI roles abroad—the scientists who have authored foundational contributions at DeepMind, OpenAI, and Google Brain, and the large Indian-origin cohort within the global semiconductor design workforce. These researchers represent a latent sovereignty asset that India has effectively exported and must now seek to repatriate. Without a critical mass of researchers trained to operate at the frontier, no quantum of compute or capital will produce frontier research.

## **6. Conclusion**

India’s current AI strategy will produce a nation of skilled AI users, LLM fine-tuners, and AI agent developers, but not the nation that invents the next novel language model architecture, the next reinforcement learning paradigm, or the next theoretical innovation in AI alignment. Our bibliometric evidence points out that India publishes prolifically but produces virtually no high-impact AI research, no frontier models, and no significant AI patents. The IndiaAI Mission, despite its ambition and scale, allocates zero rupees to fundamental AI research. Comparative

evidence from China and South Korea demonstrates that bottom-up, sustained, quality-focused R&D investment is the common precursor to genuine technological sovereignty. China’s GERD rose from 0.56 to 2.68% of GDP, and South Korea’s rose from 1.68% to 5.21% over three decades whilst India’s flatlined at 0.64%. Both these countries deliberately invested in basic research as a strategic national priority—and both now exercise a degree of AI sovereignty that India does not.

The concept of AI sovereignty goes beyond rhetoric and refers to concrete, measurable capabilities: the capacity to produce frontier LLMs under constrained conditions, the capacity to set the research agenda through high-impact publications, the capacity to own foundational intellectual property rather than license it, and the capacity to build useful AI that serves India’s billion-plus population in their own languages and cultural contexts. To be world-class in any of them ultimately requires foundational research backed by patient sovereign capital.

India has abundant human capital, robust institutional scaffolding, and ample strategic motivation to build AI sovereignty. Recent initiatives like the ANRF and the IndiaAI Mission, and the country’s world-class pool of mathematically trained researchers provide the raw material. What is missing is the policy commitment to direct these resources towards the foundational work that makes sovereignty possible, rather than the surface-layer applications that merely create the appearance of it.

In this paper, we consciously focused on AI research instead of India’s broader R&D landscape. This is because AI is a fast-moving, high-importance field dominated by a small number of countries and institutions in ways that make the sovereignty deficit especially acute and measurable. Nonetheless, the structural diagnoses advanced here—volume without impact, application-first incentives, and downstream capability without upstream research—apply with equal force to India’s broader scientific enterprise. AI is the domain where these pathologies are most visible and most consequential today. A research policy transformation that builds AI sovereignty would, by extension, begin to repair India’s broader relationship with fundamental knowledge creation: a project that is, ultimately, inseparable from the *Atmanirbhar Bharat* vision itself.

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